

Tutorial_10_Adding_a_Tool

Tutorial 10: Adding a New Tool via MCP SDK

Duration: 30 minutes | **Level:** Advanced

Overview

Create a custom tool that integrates with HelixCode's MCP server.

Prerequisites

- HelixCode running with MCP enabled
- Go 1.26.0 or later
- Understanding of JSON-RPC

Step 1: Define Tool Schema

Create `internal/tools/mycalculator/tool.go`:

```
package mycalculator

import (
    "context"
    "encoding/json"
    "fmt"
)

type CalculatorArgs struct {
    A float64 `json:"a"`
    B float64 `json:"b"`
}

func (t *CalculatorTool) Name() string { return "calculator" }
func (t *CalculatorTool) Description() string {
    return "Performs arithmetic operations on two numbers"
}

func (t *CalculatorTool) Schema() json.RawMessage {
    return json.RawMessage(`{
        "type": "object",
        "properties": {
            "a": {"type": "number"},
            "b": {"type": "number"}
        }
    `)
```

```

    },
    "required": ["a", "b"]
  })
}

```

Step 2: Implement Execute

```

func (t *CalculatorTool) Execute(ctx context.Context, args
    json.RawMessage) (string, error) {
    var parsed CalculatorArgs
    if err := json.Unmarshal(args, &parsed); err != nil {
        return "", fmt.Errorf("invalid args: %w", err)
    }
    result := parsed.A + parsed.B
    return fmt.Sprintf("%f", result), nil
}

```

Step 3: Register with MCP

In `internal/mcp/registry.go`:

```

import "dev.helix.code/tools/mycalculator"

func init() {
    RegisterTool(&mycalculator.CalculatorTool{})
}

```

Test: Start server, connect via WebSocket, send `{"jsonrpc": "2.0", "method": "tools/call", "params": {"name": "calculator", "arguments": {"a": 5, "b": 3}}}`.
 Expected response: `{"result": "8.000000"}`.

Step 4: Tool Confirmation (Optional)

Implement `Confirmable` interface for dangerous tools:

```

func (t *CalculatorTool) RequiresConfirmation() bool {
    return false // calculator is safe
}

```

Sources verified

Sources verified 2026-05-29: <https://go.dev/dl/> (go1.26.3 latest stable Go; 1.24 past support) ;
 project go.mod (root go 1.25.2, inner go 1.26) + CLAUDE.md §3.1 (PostgreSQL 15+).